

Problem D. Piece It Together

TimeLimit: 1 second
MemoryLimit: 1024 megabytes

There was a string S of lowercase English letters. Someone wrote S down n times, and then cut each of the n copies once, splitting it into a non-empty prefix and a non-empty suffix. Different copies may be cut at different positions.

You are given all $2n$ resulting pieces in an **arbitrary** order. Find any string S that could have produced them.

Input

The first line contains an integer n ($n \geq 1$). Each of the next $2n$ lines contains one piece, a non-empty string of lowercase English letters. The total length of all pieces is at most $2 \cdot 10^5$.

It is guaranteed that the pieces were produced from some string S as described.

Output

Print any string S that could have produced the given pieces.

Examples

standard input	standard output
3 ab cab abca b a bcab	abcab
2 ba b b ab	bab

Note

In the first example, the copies of **abcab** were cut as **ab|cab**, **abca|b** and **a|bcab**.

In the second example, the copies of **bab** were cut as **b|ab** and **ba|b**.